
Residual Expertise Is Not Decision Value

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Abstract

Audits of human–AI complementarity target the wrong estimand. They ask whether the human knows more than the model; deployment asks whether that knowledge would change the action. Residual predictive expertise is not residual decision value. Deployment rewards actions, and a posterior movement that does not cross a reward-induced action boundary changes nothing; a signal can be informative under log-loss and leave the deployed action unchanged. We formalize the missing quantity as *boundary regret*: the regret of holding the model’s action after observing the human. In any finite-action Bayesian decision problem, the value of consulting the human equals expected boundary regret, formally verified in Lean 4. The identity turns when humans help into a geometric question: which decision facets does the human signal move belief across? Complementarity is therefore reward-relative. A given human signal can be valuable under one reward matrix and worthless under another, the human’s knowledge unchanged; the boundary moved, not the expertise. The gap is widest precisely where asymmetric costs dominate: high-stakes clinical and operational decisions. Audits built on residual predictive expertise overstate the value of human review and route scarce review toward cases where the human is informative but the action will not move. Boundary regret is the quantity to measure.

1 Introduction

Human review is often justified by asking whether the human knows something the model does not. In clinical AI, a radiologist may catch findings that an imaging model misses, and residual-expertise audits can test whether the human signal carries predictive information beyond the model [Alur et al., 2023, 2024]. But deployment rewards actions, not predictive information. A radiologist can shift the posterior without changing the admit/discharge decision; only posterior movements that cross the reward-induced action boundary matter in practice. Residual predictive expertise and residual decision value are not the same estimand.

Complementarity is therefore reward-relative. The same human signal can matter under one reward matrix and not under another, not because the human’s expertise changed, but because the boundary moved. Human–AI complementarity is not a property of the prediction problem alone. For deployment, it also depends on the model belief, the human signal, the action set, and the reward matrix. When audits treat residual predictive expertise as residual decision value, they can spend scarce review budget on cases where the human is informative but operationally irrelevant.

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We formalize the missing estimand as *boundary regret*: the regret of keeping the model’s action after observing the human signal. For model belief b_x , model-optimal action $a_x = \arg \max_a r_a \cdot b_x$, human-updated belief $b_{x,h} = P(Y \mid x, h)$, and terminal value $V(b) = \max_a r_a \cdot b$,

$$\text{BR}^*(x, h) = V(b_{x,h}) - r_{a_x} \cdot b_{x,h}, \quad (1)$$

which is positive exactly when a rival action becomes preferred under $b_{x,h}$. In any finite-action Bayesian decision problem, the value of observing the human signal equals expected boundary regret (Theorem 2):

$$\text{VoI}(H \mid b_x) = \mathbb{E}_h[\text{BR}^*(x, h)]. \quad (2)$$

Predictive gain without a reachable boundary crossing has zero decision value for the deployed action set. We formalize in Lean 4 both the zero-regret boundary condition and the value-of-information identity above. The paper develops this criterion in four steps:

- We define boundary regret and prove, in Lean 4, that $\text{VoI}(H \mid b_x) = \mathbb{E}_h[\text{BR}^*(x, h)]$, with a geometric facet-activation criterion that characterizes when boundary regret is strictly positive (Theorems 2–3).
- We validate the plug-in estimator BR on synthetic finite-action problems, including a strong-null condition where humans carry substantial predictive signal yet boundary regret is zero everywhere: predictive gain is not sufficient for decision value.
- We show on CheXpert that residual predictive expertise and residual decision value separate as a function of reward geometry, exactly as the theory predicts: the same model, the same readers, and the same conditions yield opposite-signed correlations under different rewards.
- We show that ex ante BR improves scarce-review allocation over reward-aware margin, reward-agnostic uncertainty, residual-expertise, and learning-to-defer baselines where boundary-regret mass is reachable, and converges to reward-aware margin where it is not (Theorem 3).

2 Boundary Regret

2.1 Decision Setup

We work in finite-action Bayesian decision problems [DeGroot, 1970, Howard, 1966]. An instance with features $x \in \mathcal{X}$ has unknown state $y \in \mathcal{Y} = \{1, \dots, K\}$. The decision-maker chooses an action a from a finite set $\mathcal{A}$, $|\mathcal{A}| \geq 2$, and receives reward $r_a(y)$ specified by $R \in \mathbb{R}^{|\mathcal{A}| \times K}$. A *belief* $b \in \Delta^{K-1}$ is a probability vector over states; write $b_x = P(Y \mid x)$ for the model belief at instance x . At any belief, the terminal value is the largest expected reward available from the action set. The model-optimal action a_x attains this value under b_x . Each action has a decision region: the beliefs at which that action is optimal. Formally,

$$V(b) = \max_{a \in \mathcal{A}} r_a \cdot b, \quad a_x \in \arg \max_{a \in \mathcal{A}} r_a \cdot b_x, \quad \mathcal{R}_a = \{b : r_a \cdot b \geq r_{a'} \cdot b \ \forall a' \in \mathcal{A}\}. \quad (3)$$

Each $\mathcal{R}_a$ is a convex polytope, and the relative boundary $\mathcal{R}_a \cap \mathcal{R}_{a'}$ is a *decision facet*. The regret of action a at belief b is $\text{regret}(a, b) = V(b) - r_a \cdot b \geq 0$, with equality iff a is optimal.

2.2 Boundary-Regret Estimand

After observing the human signal h , the model belief b_x updates to $b_{x,h}$. That update has decision value only if it crosses a decision facet, making some rival action better than a_x .

Definition 1 (Boundary Regret). *For an instance x with model-optimal action a_x and human signal h , the instance-level boundary regret is the regret of keeping a_x after the belief updates from b_x to $b_{x,h}$. The ex ante boundary regret is the expected decision value of the human signal at instance x , before h is observed. Formally,*

$$\text{BR}^*(x, h) = V(b_{x,h}) - r_{a_x} \cdot b_{x,h}, \quad \text{BR}^*(x) = \mathbb{E}_{h|x}[\text{BR}^*(x, h)]. \quad (4)$$

We call $\text{BR}^*(x, h)$ the *oracle boundary regret*; BR denotes its *plug-in estimate* (§2.5).

2.3 Decision-Value Identity

The value of observing the human signal is exactly expected boundary regret. This rewrites value of information as the loss from keeping the model-only action after the posterior update.

Theorem 2 (VoI as Expected Boundary Regret). *In any finite-action Bayesian decision problem with a_x optimal at b_x , where $b_{x,h}$ is the Bayesian posterior update of b_x after observing h , the value of observing the human signal equals expected boundary regret:*

$$\text{VoI}(H \mid b_x) := \mathbb{E}_{h|x}[V(b_{x,h})] - V(b_x) = \mathbb{E}_{h|x}[\text{BR}^*(x, h)] = \text{BR}^*(x). \quad (5)$$

Proof. Use the Bayesian martingale identity: the posterior averages back to the prior. Therefore the expected reward of any fixed action is unchanged after averaging over the posterior update:

$$\mathbb{E}_{h|x}[b_{x,h}] = b_x, \quad \mathbb{E}_{h|x}[r_a \cdot b_{x,h}] = r_a \cdot b_x \quad \text{for any fixed action } a.$$

The left-hand side is the definition of value of information. Then use the identity above and the optimality of a_x at b_x . Subtract the fixed-action value from the posterior value:

$$\mathbb{E}_{h|x}[V(b_{x,h})] - V(b_x) = \mathbb{E}_{h|x}[V(b_{x,h})] - \mathbb{E}_{h|x}[r_{a_x} \cdot b_{x,h}] = \mathbb{E}_{h|x}[V(b_{x,h}) - r_{a_x} \cdot b_{x,h}]. \quad \square$$

2.4 Boundary Crossing

Boundary regret is positive exactly when at least one reachable human-updated posterior leaves the incumbent decision region. If every reachable posterior stays in $\mathcal{R}_{a_x}$, the human signal may improve prediction, but the deployed action is unchanged.

Theorem 3 (Facet Activation Bridge). *For a coherent Bayesian belief/update pair $(b_x, b_{x,h})$, there is positive ex ante boundary regret if and only if some reachable posterior strictly prefers a rival action to the model-only action. Equivalently, boundary regret is zero if and only if every reachable posterior remains in the model action’s decision region:*

$$\begin{aligned} \text{BR}^*(x) > 0 &\iff \exists h, a' \neq a_x \text{ with } P(h \mid x) > 0 \text{ and } r_{a'} \cdot b_{x,h} > r_{a_x} \cdot b_{x,h}, \\ \text{BR}^*(x) = 0 &\iff b_{x,h} \in \mathcal{R}_{a_x} \text{ for every } h \text{ with } P(h \mid x) > 0. \end{aligned} \quad (6)$$

Proof. By Theorem 2, ex ante boundary regret is the expectation of the non-negative instance-level regret. Thus $\text{BR}^*(x) > 0$ iff some reachable h has positive instance-level boundary regret. By definition, this means $V(b_{x,h}) > r_{a_x} \cdot b_{x,h}$, which holds iff some action a' achieves $r_{a'} \cdot b_{x,h} > r_{a_x} \cdot b_{x,h}$. The converse follows by contraposition: if a_x is optimal at every reachable posterior, then $\text{BR}^*(x, h) = 0$ for all h , hence $\text{BR}^*(x) = 0$. $\square$

Corollary 4 (No Crossing, No Decision Value). *If every reachable human-updated posterior $b_{x,h}$ lies in $\mathcal{R}_{a_x}$, then $\text{VoI}(H \mid b_x) = 0$, regardless of how much H improves predictions as measured by log-loss, calibration, or any proper scoring rule.*

2.5 Deployment and Estimation

Potential vs. Realized Value. $\text{BR}^*(x)$ is an information-theoretic ceiling on the per-instance decision value of human review. To realize that value, a system still needs correct elicitation, calibrated posterior updating, and operator compliance [Vaccaro et al., 2024, Bućinca et al., 2021]. Boundary regret isolates the reward-local ceiling for any deferral policy or interface design.

Estimation. The identities above are exact for ideal Bayesian beliefs: b_x and $b_{x,h}$ are a coherent prior/posterior update pair. Empirically, we replace these objects with estimates. The estimated model belief determines a_x ; a cross-fitted human-augmented model estimates the updated belief $\hat{b}_{x,h} \approx P(Y \mid x, h)$.

Thus BR is a plug-in estimate of boundary regret, not a new theoretical identity. For ex ante review allocation, we also estimate $P(h \mid x)$ before human review and compute

$$\hat{\mathbb{E}}_{h|x}[V(\hat{b}_{x,h}) - r_{a_x} \cdot \hat{b}_{x,h}]. \quad (7)$$

Calibration affects how accurately this plug-in score recovers true boundary regret; Section 3.1 studies this sensitivity. Section 3 also details how this quantity is estimated for each experiment.

3 Experiments

We have three sequential experiments. Section 3.1 validates the boundary-regret estimator in synthetic settings with closed-form ground truth. Section 3.2 tests whether residual predictive expertise and residual decision value diverge on a real human–AI dataset across reward structures. Section 3.3 asks whether an ex ante boundary-regret score allocates scarce human review better than standard baselines under asymmetric rewards. All three experiments run end-to-end on standard CPU workers.

3.1 Experiment 1: Estimator and Geometry Validation

Question. Does the plug-in estimator recover known boundary regret, and remain zero when a safe action dominates the reachable belief region?

Setup. We simulate finite-state Bayesian decision problems with known prior, reward matrix, and channels $P(m | y), P(h | y) (M \perp H | Y)$. For each sampled (y, m, h) we compute the exact posteriors $b_m = P(y | m)$ and $b_{m,h} = P(y | m, h)$, the model-only action a_m , and instance-level boundary regret

$$\text{BR}^*(m, h) = V(b_{m,h}) - r_{a_m} \cdot b_{m,h}.$$

The estimator does not see the true channels: it estimates $\hat{P}(m | y), \hat{P}(h | y), \hat{p}(y)$ from finite data via 5-fold cross-fitting with Laplace smoothing, then computes plug-in BR on held-out instances ($N = 10,000$ per condition).

Conditions. The grid varies $G \in \{2, 3, 5\}$ decision groups and $K \in \{G, 2G\}$ latent states, covering the minimal boundary case ($G = 2$), the smallest genuinely multi-action case ($G = 3$), and a moderate stress test ($G = 5$); $K = 2G$ checks that estimation does not degrade when multiple latent states share a decision group. Three roles: C1 is a positive control (actions reward state groups, human signal crosses boundaries), C2 a weak null (prior deep inside one decision region, weak human channel), and C3 a strong null (highly predictive human signal but a safe action dominates every reachable posterior). For C3 we set $\tau = (1 + U)/2$, where U is the maximum reachable posterior mass on any decision group under the exact channels; $\tau > U$ guarantees no reachable posterior can make a non-safe action optimal while the human signal retains its predictive accuracy.

Results. Table 1 reports metrics averaged across the six configurations; rank correlation is reported only where BR^* varies (in both nulls $\text{BR}^* \equiv 0$, so we report false-positive rate instead).

Table 1: Synthetic validation of BR averaged over $G \in \{2, 3, 5\}, K \in \{G, 2G\}$. C3’s ΔLL (0.595) exceeds C1’s because the human signal is strongly predictive of group membership, yet no posterior crosses the certified threshold. ΔLL : log-loss improvement from b_m to $b_{m,h}$.

Condition	Role	$\rho(\text{BR}, \text{BR}^*)$	FPR	% $\text{BR}^* > 0$	% $\text{BR} > 0$	ΔLL
C1	Positive control	0.901	0.000	37.9	37.9	0.408
C2	Weak null	n/a	0.000	0.0	0.0	0.019
C3	Strong null	n/a	0.000	0.0	0.0	0.595

The estimator separates predictive value from decision value. In C1, BR ranks true boundary regret well ($\rho = 0.901$) across the 37.9% of instances where crossings occur (per-configuration details in Appendix B); FPR is zero in both nulls. C3 is the critical case: the human signal achieves the highest log-loss improvement of any condition ($\Delta\text{LL} = 0.595$), yet BR assigns zero boundary regret to every instance because no posterior crosses the certified threshold. High predictive accuracy on task-relevant features does not imply decision value when a safe action dominates the reachable belief region.

3.2 Experiment 2: Residual Expertise Does Not Equal Decision Value

Question. Does the correlation between residual predictive expertise and boundary regret invert across reward regimes on the same model, readers, and data?

Setup. We pair a pretrained DenseNet-121 chest X-ray classifier [Huang et al., 2017, Cohen et al., 2022] with five board-certified radiologists of the CheXpert reader study [Irvin et al., 2019] across five official competition conditions (Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion), yielding $N = 25$ reader–task pairs on 500 frontal radiographs. For each pair, the ground-truth label y is the majority vote of the *remaining four* readers, which eliminates structural h – y contamination and ensures $y \perp h$. Model beliefs b_x are temperature-scaled per pair on a 30% calibration split; evaluation uses the remaining 70% (≈ 350 images per pair). The human-augmented posterior $\hat{b}_{x,h} \approx P(Y = 1 \mid x, h)$ is estimated via 5-fold cross-fitted logistic regression on features $[\text{logit}(b_x), h, h \cdot \text{logit}(b_x)]$. Spearman correlations between residual predictive expertise (log-loss gain over b_x) and BR are reported across the 25 pairs under each reward regime; label protocol details in Appendix C.

Reward Regimes. We evaluate three reward matrices $r_{a,y}$, each placing the decision threshold $r_{0,0}/(r_{0,0} + r_{1,1})$ [Pauker and Kassirer, 1980] in a distinct region of the model belief distribution. R1 is the identity ($r_{0,0} = r_{1,1} = 1$, errors 0), threshold 1/2, in the bulk. R2 ($r_{0,0} = 1, r_{1,1} = 5$, errors 0) penalises false negatives five times as heavily as false positives, threshold 1/6, below the bulk. R3 mirrors R2 ($r_{0,0} = 5, r_{1,1} = 1$), threshold 5/6, in the sparse upper tail. R3 is the critical negative control: Theorem 3 predicts that when the threshold sits in the tail, facet crossings become rare, boundary-regret mass collapses to near zero, and residual predictive expertise decouples from decision value regardless of how much the reader improves the model’s log-loss.

Table 2: Boundary-regret diagnostics on $N = 25$ reader–task pairs: actionable BR mass collapses from R1 to R3 and BR concentrates near the active facet in all regimes. n_v : pairs where $\text{BR} \neq 0$; $\bar{\rho}_{\text{facet}}$: mean Spearman between facet distance and BR.

Reward	Role	$\bar{\rho}_{\text{LL}}$	$\rho > 0/n_v$	$\bar{\rho}_{\text{facet}}$	Top-decile zero-BR	n_v
R1 (thr. 1/2)	Primary result	+0.477	25/25	−0.599	11.2%	25
R2 (thr. 1/6)	Intermediate	−0.051	15/25	−0.866	28.3%	25
R3 (thr. 5/6)	Negative control	−0.112	0/7	−0.221	97.4%	7

Reward-Conditional Dissociation. The correlation between residual predictive expertise (log-loss gain over the model alone) and boundary regret inverts with the reward (Table 2): $\bar{\rho}_{\text{LL}} = +0.477$ under R1 (25/25 pairs positive), -0.051 under R2 (15/25), and -0.112 under R3 (0/7; sign test $p = 0.008$, with the remaining 18 of 25 pairs structurally $\text{BR} \equiv 0$). Same model, same readers, same data; only the reward changes.

The mechanism is geometric. Under R1, threshold 1/2 sits in a dense region of model beliefs, so a single reader update routinely crosses a facet and predictive expertise aligns with decision value. Under R3, the false-positive facet sits in the tail and no posterior update reaches it: only 7 of 25 pairs carry any non-zero BR, and 97.4% of top-expertise-decile cases (90th percentile of model–reader disagreement $1 - b_x[h]$) have $\text{BR} = 0$. The R3 correlation $\bar{\rho} = -0.112$ on these 7 pairs is directionally consistent with the geometric prediction; the collapse of n_v from 25 to 7 is the direct confirmation that boundary-regret mass has evacuated. R2 is intermediate (28.3% top-decile zero-BR). Facet distance correlates negatively with BR under every reward ($-0.599, -0.866, -0.221$ for R1, R2, R3): boundary regret concentrates near the active facet regardless of how predictive the reader is (Theorem 3). R2’s stronger concentration follows from the same geometry: with the threshold at 1/6 nearly all decisions default to treat, so the rare facet crossings come precisely from beliefs already near 1/6. R1’s centrally located threshold produces no such concentration. We call this *reward-conditional dissociation*: a consequence of reward geometry, not a calibration failure.

Appendix D shows the inversion and dissociation hold across in- and off-domain pretrained checkpoints (varying the belief geometry) and readers not involved in ground-truth labelling. As a theory-confirming negative control, Appendix E evaluates CIFAR-10H: on highly accurate classifiers whose softmax probabilities rarely straddle a reward facet, $\bar{\rho}$ collapses to near-zero across all rewards ($+0.015$ under R1 vs. $+0.477$ on CheXpert R1), confirming that the dissociation signal depends on the model’s beliefs straddling the active facet, exactly as Theorem 3 predicts.

Implication for Predictive Audits. A residual-expertise audit in the style of Alur et al. [2023, 2024] would find substantial predictive value here (mean log-loss gain 0.387 nats across all 25

pairs) and recommend deploying the readers. Under R3, 97.4% of those same cases carry zero decision value. The *estimand shift* from predictive to decisional complementarity is not a refinement; it changes the sign of the deployment recommendation. If this regime-conditional inversion is theorem-predicted, an ex ante BR score should inherit it: dominating reward-aware geometric proximity where boundary-regret mass is reachable, converging with it where mass is absent. Section 3.3 tests this directly.

3.3 Experiment 3: Review Allocation under Scarce Human Input

Question. Given a fixed human-review budget, can an ex ante boundary-regret score identify the cases where reader input will improve realized decision utility, rather than improve prediction or flag model uncertainty? The prediction is regime-conditional: it should beat reward-aware but human-blind boundary proximity when reader updates can cross reward-relevant decision boundaries, and converge to it when such crossings are sparse or absent.

Setup. We use the same 25 reader–task pairs as in Section 3.2. For each case, the model-only belief b_x induces a model-only action a_x . For each review budget $q \in \{5\%, 10\%, 20\%, 50\%\}$, a policy ranks cases using only pre-review information. Reviewed cases observe the reader response h and take the human-updated action $\arg \max_a r_a \cdot \hat{b}_{x,h}$; unreviewed cases retain a_x . Utility gain is the held-out realized reward improvement over this no-review baseline. All learned policy scores are 5-fold cross-fitted on the same fold structure used to estimate $\hat{b}_{x,h}$, so review priority is always computed out of fold.

Policies. The proposed policy, BR, is an ex ante boundary-regret score: it predicts possible reader responses from x , predicts the corresponding human-updated posterior, and scores a case by the expected decision value of that update under the current reward. We compare it against six baselines. **Margin** is a reward-aware distance to the active decision facet, our primary comparison baseline: it captures reward geometry without estimating the human signal $P(h \mid x)$. **Residual** is a residual-expertise score, the deployable analogue of a predictive audit [Alur et al., 2023]. **Entropy** is reward-blind model uncertainty. These policies span two axes: Margin is reward-aware but human-blind, Residual is human-aware but reward-blind, Entropy is blind to both, and BR is both reward-aware and human-aware. **L2D** is the learned full-coverage competitor: like BR it is both reward-aware and human-aware, but reaches that coverage end-to-end through a cross-fitted deferral classifier with reward-sensitive labels $\mathbb{1}[R[a_{x,h}, y] > R[a_x, y]]$, in the spirit of [Mozannar and Sontag, 2020, Mao et al., 2023], rather than through the reward’s geometry and the reader-channel posterior. **Random** is the uninformed baseline. **Oracle** is the hindsight upper bound using realized labels.

Regime-Conditional Allocation. We evaluate all policies on the same 25 reader–task pairs under each reward matrix. Table 3 tests whether the dissociation in Section 3.2 becomes actionable under a scarce review budget. It does, but only where reward geometry leaves actionable mass reachable: BR beats Margin at every R1 budget, leads or ties across most R2 budgets, and has no systematic advantage in the R3 negative control. The paired wins, losses, and ties isolate when the human-aware term adds value beyond reward geometry.

Dense actionable mass (R1). BR wins decisively at every budget (sign-test $p < 0.001$); decisive-win rates over Margin are 91%, 100%, 96%, 100% (tallies 20/2/3, 24/0/1, 22/1/2, 25/0/0). With actionable mass dense in the belief distribution, the reader channel adds signal beyond reward-aware geometric proximity: Margin finds cases near the active facet; BR identifies which of those the reader can move profitably. L2D matches BR’s nominal coverage but falls below Random at $q \leq 20\%$ (+0.0152 vs. +0.0184 at $q = 5\%$): the structured decomposition is what makes that coverage usable.

Sparse actionable mass (R2). The low threshold places most cases on the positive action; sparse boundary-regret mass concentrates near the active facet (exactly where Margin sorts), making Margin a strong baseline. All deployable reward-aware policies obtain much larger absolute gains than under R3. BR leads or ties Margin at every budget up to $q = 20\%$ with no paired losses (3/0/22, 3/0/22, 5/2/18); the $q = 50\%$ row is essentially a tie (7/9/9, mean difference 0.005). The human-aware term adds value where actionable mass exists and is harmless where mass is sparse.

Table 3: Ex ante review allocation under scarce human input. Entries report mean held-out utility gain over no review on $N = 25$ reader-task pairs. BR beats reward-aware Margin where actionable mass is dense (R1), leads or ties where mass is sparse (R2), and has no systematic edge where mass is absent (R3, negative control). W/L/T gives paired wins/losses/ties of BR against Margin; bold marks the best deployable policy in each row (Oracle excluded).

Reward	q	BR	Margin	Entropy	Residual	L2D	Random	Oracle	W/L/T
R1	5%	+0.0459	+0.0395	+0.0395	+0.0209	+0.0152	+0.0184	+0.0514	20/2/3
R1	10%	+0.0883	+0.0743	+0.0743	+0.0455	+0.0289	+0.0392	+0.1000	24/0/1
R1	20%	+0.1638	+0.1438	+0.1438	+0.0915	+0.0697	+0.0761	+0.2000	22/1/2
R1	50%	+0.3325	+0.2797	+0.2797	+0.2544	+0.2485	+0.2022	+0.4495	25/0/0
R2	5%	+0.0514	+0.0490	+0.0429	+0.0330	+0.0477	+0.0313	+0.0514	3/0/22
R2	10%	+0.0999	+0.0965	+0.0768	+0.0610	+0.0946	+0.0534	+0.1000	3/0/22
R2	20%	+0.1914	+0.1889	+0.1486	+0.1286	+0.1879	+0.1089	+0.2000	5/2/18
R2	50%	+0.4113	+0.4161	+0.2955	+0.3069	+0.4134	+0.2664	+0.4865	7/9/9
R3	5%	+0.0029	+0.0027	+0.0000	+0.0000	-0.0008	+0.0006	+0.0072	1/4/20
R3	10%	+0.0025	+0.0031	+0.0000	+0.0000	+0.0008	+0.0007	+0.0072	0/4/21
R3	20%	+0.0025	+0.0029	+0.0000	+0.0000	+0.0017	+0.0006	+0.0072	1/4/20
R3	50%	+0.0027	+0.0029	+0.0000	+0.0000	+0.0023	+0.0021	+0.0072	1/2/22

No actionable mass (R3). The active facet sits in the sparse upper tail of model beliefs; eighteen of twenty-five pairs have $BR \equiv 0$ across the evaluation set. Allocation gains are correspondingly tiny: BR and Margin differ by less than 0.0006 at every budget, with only three to five decisive pairs out of 25. This is the no-actionable-mass prediction of Theorem 3, not an allocation failure.

Human Channel versus Geometric Proximity. Reward-aware distance to the active facet is a strong baseline: it already identifies cases near the decision boundary. The decisive-win rates over Margin under R1 show that BR’s estimate of $P(h \mid x)$ and $P(Y \mid x, h)$ add reliable signal on top of that geometric distance where actionable mass is dense. Where boundary-regret mass is sparse (R2) or absent (R3), BR and Margin converge, both outcomes predicted by Theorem 3. Residual and Entropy clarify why both axes matter: human information without reward geometry, or model uncertainty without either reward geometry or the reader channel, allocates review less efficiently. L2D makes the opposite case: nominal coverage of both axes is not enough on its own; the structured decomposition is what makes that coverage usable at this sample size. The reward-aware/human-aware policy adds value exactly when both components are needed; it does not manufacture decision value when the reachable belief region carries none.

4 Related Work

Four literatures bear most directly on boundary regret: residual human expertise, deferral and selective prediction, realized human-AI collaboration, and the decision-theoretic value of information.

Residual Human Expertise Audits. A line of work asks whether human judgments carry predictive information an algorithm has missed. Alur et al. [2023] test for residual expertise using conditional covariance, log-loss improvement, and conditional-independence checks; Alur et al. [2024] give an algorithmic-indistinguishability account of the same question. Kleinberg et al. [2018] formalize the selective-labels problem: when human labels are observed only on reviewed cases, estimating how outcomes depend on both the instance and the human judgment requires accounting for non-random review. Guo et al. [2025] push the question to the feature level via ILIV-SHAP, attributing each input feature’s contribution to the agent-complementary VoI of an AI prediction. What these papers ask is whether humans carry residual predictive information, and which features drive it. We ask something narrower and decision-relevant: whether that information actually changes the reward-optimal action. Residual expertise can sit comfortably alongside zero residual decision value, and a feature-level decomposition does not tell you whether any of its predictive contributions cross a reward-induced decision facet.

Deferral, Selective Prediction, and Reject Options. Learning-to-defer methods train systems to choose between acting automatically and handing an instance to a human expert [Madras et al., 2018, Mozannar and Sontag, 2020, Hemmer et al., 2023, Mao et al., 2023]. Selective-prediction and reject-

option methods address the related problem of when a classifier should abstain under a specified loss [Ramaswamy et al., 2018, Franc et al., 2023], and cost-sensitive variants fold asymmetric penalties into the deferral objective [Alves et al., 2024]. All of these decide *when* to defer or abstain given a loss, an expert channel, and a collaboration protocol. The question we want to answer comes earlier: whether the human signal can move the deployed action across a reward-induced boundary at all, before any such protocol is on the table.

Realized Human–AI Collaboration. A separate body of work studies when teams actually achieve complementarity in deployment. Vaccaro et al. [2024] synthesize evidence that human–AI combinations often fail to outperform the better standalone agent, and Peng et al. [2025] prove a structural counterpart: any deterministic strategy combining calibrated forecasts that does not always defer to one agent will sometimes underperform the least accurate agent. Schoeffer et al. [2024] show that explanations can change reliance without necessarily helping users identify correct advice, while Guo et al. [2024] propose a decision-theoretic framework that decomposes observed deployment loss into mis-reliance and signal-perception components. McLaughlin and Spiess [2024] model recommendation design for realized complementarity, and Hemmer et al. [2025] organize the empirical evidence on when complementarity appears. A related critical strand argues that complementarity also turns on action-set design, evaluation metrics, and interaction design [Straitouri et al., 2025]. That work is about teams once they are deployed: whether they realize value, what protocols are achievable in principle, how observed loss decomposes after the fact. Our question runs upstream of all of that. Is reward-local decision value in the joint distribution to begin with, before strategy choice, behavioral frictions, interfaces, and incentives have any chance to surface it or hide it?

Value of Information, Decision Geometry, and Decision-Focused Learning. Conceptually we are closest to value-of-information theory and decision geometry. Blackwell [1951] orders information structures by decision value. Radner and Stiglitz [1984] and de Lara and Gossner [2020] characterize when information value vanishes because posteriors stay inside a single action region, and Kamenica and Gentzkow [2011] expresses information value through convex value functions and induced decision regions. Decision-focused learning makes the adjacent algorithmic point that prediction should be trained for downstream decisions rather than generic accuracy [Elmachoub and Grigas, 2022, Wilder et al., 2019], and decision calibration asks when predictions are indistinguishable to downstream best-response rules [Zhao et al., 2021]. In clinical statistics, decision curve analysis [Vickers and Elkin, 2006] evaluates predictive models under threshold-encoded reward rather than generic accuracy, plotting net benefit across a range of clinician decision thresholds; that work established the principle that model utility should be assessed under its downstream loss. All of this is about decisions in the abstract, or about how a single model performs under reward. Our move is to take the same geometry and apply it as a human–AI audit: a reward-local check on whether the human signal has decision value for the action actually being deployed, with the underlying VoI identity verified in Lean.

5 Conclusion

The audit question for human–AI complementarity is not whether the human carries information the model lacks but whether that information would change the reward-relevant action. We named the missing estimand boundary regret and proved that it equals the value of consulting the human in any finite-action Bayesian decision problem; that quantity is strictly positive precisely when some reachable posterior crosses a decision facet. The plug-in score this identity yields behaves as the bridge predicts: it allocates scarce review where decision-relevant mass is reachable and converges on reward-aware margin where it is not. No residual-expertise audit produces either behavior.

Limitations. Boundary regret characterizes the reward-local decision value latent in the joint distribution of (x, h, y) . Whether a deployed workflow elicits and uses that value depends on the elicitation interface and on behavioral frictions [Vaccaro et al., 2024, Bućinca et al., 2021] the estimand is not designed to model. The deployed score is a plug-in estimate of an exact identity; calibration of the model belief and the human-augmented posterior controls the gap between the two.

Broader impact. Routing scarce human review to cases that change the reward-relevant action raises operational efficiency in high-stakes settings such as clinical reading and triage. But the same machinery, fed a misspecified or contested reward, will under-allocate review for populations whose decision facets it does not reach. Any operational use of BR should pair the score with reward elicitation and a subgroup audit of which populations it removes human review from.

Outlook. Boundary regret does not single out the human as the information source. The same identity prices any candidate channel (a second-opinion reader, an additional diagnostic test, a more expensive feature) by what it would change about the reward-relevant action. A reward-faithful training signal for learning-to-defer follows: replace accuracy-based deferral labels with the regret of not consulting, and the construction extends to multi-expert and sequential-elicitation policies. A prospective deployment routing review on BR under operational rewards is the direct test of how much latent decision value a real workflow can extract. Complementarity is reward-relative. A signal can be informative under any proper scoring rule and leave the deployed action unchanged. The audit that ignores the boundary measures the wrong thing.

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A Lean 4 Proof Artifacts

All formal results used in the paper are verified in Lean 4 using Mathlib. The development contains five modules: `Basics.lean`, `Regret.lean`, `JensenGap.lean`, `FacetActivation.lean`, and the umbrella import `ResidualExpertiseNotDecisionValue.lean`. Table 4 maps the paper’s claims to Lean declarations.

Table 4: Map from paper statements to Lean 4 declarations.

Paper	Type	Lean declaration	Module
$V(b) = \max_a r_a \cdot b$	Def	<code>terminalValue</code>	Regret
$\text{regret}(a, b) \geq 0$	Lem	<code>regret_nonneg</code>	Regret
$\text{regret} = 0 \Leftrightarrow a \text{ optimal}$	Lem	<code>regret_eq_zero_iff</code>	Regret
$\mathbb{E}[b_{\text{post}}] = b$	Lem	<code>prior_eq_weighted_posterior</code>	Basics
Jensen gap ≥ 0	Thm	<code>jensenGap_nonneg</code>	JensenGap
Thm 2 ($\text{VoI} = \mathbb{E}[\text{BR}^*]$)	Thm	<code>jensenGap_eq_weighted_regret</code>	FacetActivation
Thm 3	Thm	<code>facet_activation_bridge</code>	FacetActivation

Primitives. `Basics.lean` defines `Belief`, `ObsKernel`, `marginalProb`, and `posterior`, and proves the posterior martingale identity `prior_eq_weighted_posterior`. `Regret.lean` defines `terminalValue` and `regret` and proves non-negativity and the zero-regret optimality characterization. `JensenGap.lean` defines Jensen gaps over posterior lotteries and proves non-negativity for belief-convex functions.

Paper Theorems. `FacetActivation.lean` contains the two formal claims used in the text. The weighted-regret identity `jensenGap_eq_weighted_regret` rewrites the value of information as expected regret of retaining the prior-optimal action after the posterior update. The bridge theorem `facet_activation_bridge` states that this expected regret is strictly positive exactly when some reachable posterior strictly prefers a rival action.

B Synthetic Validation: Per-Configuration Results

Table 5 reports per-configuration results across the three estimator-validation conditions of Section 3.1. False-positive rate is zero in every cell: across all 18 configurations the plug-in estimator never assigned positive boundary regret to an instance with $\text{BR}^*(m, h) = 0$. Under the positive control C1, $\rho(\text{BR}, \text{BR}^*)$ degrades smoothly as the action/state space grows (G from 2 to 5, K from 2 to 10, giving $0.969 \rightarrow 0.745$), reflecting harder channel estimation, not a failure of the identity. Under the strong null C3, the human signal is by construction strongly predictive (ΔLL up to 0.738), yet the estimator continues to assign zero boundary regret to every instance, the geometric prediction of Theorem 3.

C CheXpert Protocol

Reader Study and Label Construction. The five board-certified radiologists (bc1, bc2, bc3, bc5, bc7) provided binary disease-presence ratings under the official CheXpert five-condition protocol (Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion) on the 500-image frontal-view validation set. For each (condition, ground-truth reader) pair, the held-out label y is computed by *leave-one-out majority vote*: for reader r_i , $y = \mathbb{1}[\sum_{j \neq i} v_j \geq 3]$, where $v_j \in \{0, 1\}$ are the remaining four ground-truth readers’ votes. This construction enforces $y \perp h$ structurally: the test reader is never a voter in their own ground truth. The three benchmark readers (bc4, bc6, bc8) were not voters in the original study, so their pairs use all five ground-truth votes (equivalent to the official majority).

Model Belief and Calibration. Model probabilities b_x^{raw} are produced by the `torchxrayvision` `DenseNet-121` [Huang et al., 2017, Cohen et al., 2022] with the standard X-ray preprocessing pipeline (8-bit normalisation, centre crop, 224×224 resize). The primary experiment uses the in-domain checkpoint `densenet121-res224-chex`; the robustness analyses below use all six publicly

Table 5: Per-configuration results for synthetic estimator validation. G : decision groups; K : latent states; $|A|$: actions in the reward matrix. FPR is zero in every cell and is omitted; ρ is omitted when true boundary regret is constant zero (C2, C3).

Condition.	G	K	$ A $	$\rho(\text{BR}, \text{BR}^*)$	% BR* > 0	ΔLL
C1	2	2	2	0.969	35.5	0.226
C1	2	4	2	0.984	36.4	0.230
C1	3	3	3	0.941	37.9	0.403
C1	3	6	3	0.903	38.2	0.420
C1	5	5	5	0.862	39.8	0.577
C1	5	10	5	0.745	39.8	0.592
C2	2	2	2	n/a	0.0	0.006
C2	2	4	2	n/a	0.0	0.006
C2	3	3	3	n/a	0.0	0.015
C2	3	6	3	n/a	0.0	0.019
C2	5	5	5	n/a	0.0	0.035
C2	5	10	5	n/a	0.0	0.033
C3	2	2	3	n/a	0.0	0.447
C3	2	4	3	n/a	0.0	0.458
C3	3	3	4	n/a	0.0	0.592
C3	3	6	4	n/a	0.0	0.613
C3	5	5	6	n/a	0.0	0.723
C3	5	10	6	n/a	0.0	0.738

available checkpoints. For each (condition, reader) pair we apply per-pair logit-space temperature scaling: a stratified 30% calibration split (seed 42), with scalar T found by NLL minimisation in $[0.1, 10]$, gives the calibrated belief $b_x = \sigma(\text{logit}(b_x^{\text{raw}})/T)$.

Human-Augmented Posterior. On the remaining 70% evaluation split, we estimate $\hat{b}_{x,h} \approx P(Y = 1 \mid x, h)$ by 5-fold cross-fitted logistic regression on features $[\text{logit}(b_x), h, h \cdot \text{logit}(b_x)]$, with folds stratified on y (StratifiedKFold). The leave-one-out label construction makes $y \perp h$ structural; cross-fitting additionally makes the augmented-belief estimator independent of the held-out label within each fold, so plug-in BR is not contaminated by within-fold overfitting.

Evaluation. For each pair, all per-instance quantities (BR, log-loss gain, facet distance) are computed on the held-out fold of each instance and aggregated across folds. Pair-level summaries (Spearman ρ between log-loss gain and BR, top-decile zero-mass fraction) are then aggregated across the 25 pairs. Pairs with $\text{BR} \equiv 0$ across the entire evaluation set are excluded from the ρ computation; we report the count of valid pairs n_v alongside.

D Reward-Conditional Dissociation: Robustness

Table 6 extends the dissociation analysis of Section 3.2 to all six publicly available DenseNet checkpoints and all eight CheXpert radiologists ($6 \times 8 \times 5 = 240$ reader-task pairs). The ordering $\text{R1} \gg \text{R2} \geq \text{R3}$ holds on every checkpoint individually: R1 is positive on every checkpoint (+0.373 to +0.530), R3 ranges from clearly negative (in-domain, -0.117 to -0.040) to essentially zero (off-domain NIH, +0.007), and R1 strictly dominates R2 throughout, with the gap compressing on off-domain pretraining. Off-domain pretraining (NIH, PadChest) attenuates the R1 signal and inflates R2, consistent with calibration drift moving belief mass toward the lower R2 facet at 1/6. The direction of the dissociation, not its absolute magnitude, is the theory-relevant prediction.

E CIFAR-10H: Theory-Confirming Negative Control

CIFAR-10H evaluates seven pretrained CIFAR-10 classifiers against 50 human annotators across 10 one-vs-rest binary tasks (3,500 reader-task pairs). Pretrained CIFAR-10 classifiers are highly accurate; their softmax probabilities rarely straddle a decision boundary for a given annotator’s image subset. Theorem 3 therefore predicts near-zero actionable boundary-regret mass everywhere.

Table 6: Reward-conditional dissociation across 6 DenseNet checkpoints $\times$ 8 readers $\times$ 5 conditions = 240 reader–task pairs. Entries report $\bar{\rho}_{LL}$, the mean per-pair Spearman correlation between log-loss gain and BR, under each reward (R1: thr. 1/2; R2: thr. 1/6; R3: thr. 5/6). Bold marks the aggregate dissociation pattern across all 240 pairs.

Checkpoint	R1 $\bar{\rho}_{LL}$	R2 $\bar{\rho}_{LL}$	R3 $\bar{\rho}_{LL}$
<i>In-domain pretraining</i>			
densenet121-res224-chex	+0.489	−0.008	−0.117
densenet121-res224-mimic_ch	+0.496	+0.197	−0.040
densenet121-res224-mimic_nb	+0.495	+0.210	−0.042
densenet121-res224-all	+0.530	+0.151	−0.082
<i>Off-domain pretraining</i>			
densenet121-res224-nih	+0.373	+0.313	+0.007
densenet121-res224-pc	+0.435	+0.313	−0.050
All 240 pairs	+0.470	+0.196	−0.056

Table 7: CIFAR-10H negative control: mean Spearman $\bar{\rho}$ between log-loss gain and BR across 3,500 reader–task pairs.

Reward	$\bar{\rho}$ (valid)	$\bar{\rho}$ (all)	N valid	Top-decile BR = 0
R1 (thr. 0.500)	+0.015	+0.007	1,668	96.2%
R2 (thr. 0.167)	+0.044	+0.024	1,905	94.5%
R3 (thr. 0.833)	−0.008	−0.004	1,632	96.9%

The R1 dissociation signal ($\bar{\rho} = +0.015$) is $\sim 32\times$ weaker than the primary CheXpert R1 result ($\bar{\rho} = +0.477$ across the 25 in-domain reader–task pairs; Section 3.2) and $\sim 30\times$ weaker than the 240-pair robustness average (+0.470; Table 6). The collapse is the predicted consequence of dense actionable mass being absent, not a methods failure.

F Allocation Advantage: Robustness

Table 8 extends the allocation experiment of Section 3.3 to all six DenseNet checkpoints and all eight CheXpert radiologists (240 reader–task pairs). Three patterns reproduce the main result. Under R1 (dense actionable mass), BR dominates Margin at every budget ($p < 0.0001$ for $q \geq 10\%$), confirming the advantage is not an artefact of the primary checkpoint. Under R2 (sparse but reachable mass), BR now also beats Margin at $q \in \{5\%, 10\%, 20\%\}$ ($p < 0.0001$), an effect not visible on the 25-pair primary benchmark; the dissociation table above explains why: off-domain checkpoints concentrate more belief mass near the R2 facet at 1/6, providing actionable mass that the in-domain checkpoint lacks. Under R3 (no actionable mass), BR and Margin remain economically indistinguishable, with no systematic advantage in either direction (sign-test $p \geq 0.08$ at every budget; mean differences ≤ 0.0012 at $q = 5\%$ and ≤ 0.0004 for $q \geq 10\%$), reproducing the no-mass prediction.

Table 8: Allocation robustness across 6 DenseNet checkpoints $\times$ 8 readers $\times$ 5 conditions = 240 reader-task pairs. Entries report mean held-out utility gain over no review. W/L/T gives paired wins/losses/ties of BR against Margin; bold marks the best deployable policy in each row (Oracle excluded).

Reward	q	BR	Margin	Entropy	Residual	L2D	Random	Oracle	W/L/T
R1	5%	+0.0407	+0.0403	+0.0403	+0.0192	+0.0101	+0.0195	+0.0513	106/74/60
R1	10%	+0.0787	+0.0763	+0.0763	+0.0413	+0.0247	+0.0393	+0.0996	134/71/35
R1	20%	+0.1534	+0.1444	+0.1444	+0.0939	+0.0647	+0.0788	+0.1931	167/51/22
R1	50%	+0.3160	+0.2936	+0.2936	+0.2653	+0.2267	+0.1947	+0.4132	198/25/17
R2	5%	+0.0429	+0.0350	+0.0368	+0.0186	+0.0258	+0.0215	+0.0604	112/25/103
R2	10%	+0.0826	+0.0665	+0.0685	+0.0385	+0.0541	+0.0426	+0.1078	121/23/96
R2	20%	+0.1557	+0.1252	+0.1276	+0.0870	+0.1123	+0.0849	+0.2013	132/22/86
R2	50%	+0.3213	+0.3052	+0.2682	+0.2430	+0.2826	+0.2079	+0.4399	110/98/32
R3	5%	+0.0042	+0.0030	+0.0001	+0.0008	+0.0025	+0.0002	+0.0115	22/31/187
R3	10%	+0.0052	+0.0048	+0.0001	+0.0013	+0.0043	+0.0003	+0.0120	18/30/192
R3	20%	+0.0055	+0.0056	+0.0001	+0.0020	+0.0049	+0.0005	+0.0120	13/23/204
R3	50%	+0.0057	+0.0057	+0.0003	+0.0025	+0.0050	+0.0017	+0.0120	8/18/214

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